

Doppler-free three-photon coherence in Doppler-broadened diamond-type atomic system

YOON-SEOK LEE AND HAN SEB MOON*

Department of Physics, Pusan National University, Geumjeong-Gu, Busan 46241, South Korea

**hsmoon@pusan.ac.kr*

Abstract: We investigated the relationship between three-photon electromagnetically induced absorption (TPEIA) and four-wave mixing (FWM) in the $5S_{1/2}$ – $5P_{3/2}$ – $5D_{5/2}$ transition of ^{87}Rb atoms. When the driving field was additionally coupled to the ground and intermediate states of a lower V-type configuration in a typical ladder-type electromagnetically induced transparency (EIT) system, Doppler-free TPEIA and FWM spectra were simultaneously observed and then analyzed from the perspective of multi-photon atomic coherence. Comparing the TPEIA and FWM spectra according to the laser frequency detuning and laser intensity, we found that the enhanced TPEIA signal is strongly correlated with the generated FWM light. Analytical and numerical calculations for the analysis of the relationship are presented.

© 2017 Optical Society of America

OCIS codes: (270.1670) Coherent optical effects; (020.1670) Coherent optical effects; (190.4223) Nonlinear wave mixing.

References and Links

1. K.-J. Boller, A. Imamoglu, and S. E. Harris, "Observation of electromagnetically induced transparency," *Phys. Rev. Lett.* **66**(20), 2593–2596 (1991).
2. S. E. Harris, "Electromagnetically Induced Transparency," *Phys. Today* **50**(7), 36–42 (1997).
3. M. Xiao, Y. Li, S. Jin, and J. Gea-Banacloche, "Measurement of dispersive properties of electromagnetically induced transparency in rubidium atoms," *Phys. Rev. Lett.* **74**(5), 666–669 (1995).
4. D. J. Fulton, S. Shepherd, R. R. Moseley, B. D. Sinclair, and M. H. Dunn, "Continuous-wave electromagnetically induced transparency: a comparison of V, Lambda, and cascade systems," *Phys. Rev. A* **52**(3), 2302–2311 (1995).
5. H. Schmidt and A. Imamoglu, "Giant Kerr nonlinearities obtained by electromagnetically induced transparency," *Opt. Lett.* **21**(23), 1936–1938 (1996).
6. A. M. Akulshin, S. Barreiro, and A. Lezama, "Electromagnetically induced absorption and transparency due to resonant two-field excitation of quasidegenerate levels in Rb vapor," *Phys. Rev. A* **57**(4), 2996–3002 (1998).
7. A. Lezama, S. Barreiro, and A. M. Akulshin, "Electromagnetically induced absorption," *Phys. Rev. A* **59**(6), 4732–4735 (1999).
8. M. D. Lukin, S. F. Yelin, M. Fleischhauer, and M. O. Scully, "Quantum interference effects induced by interacting dark resonances," *Phys. Rev. A* **60**(4), 3225–3228 (1999).
9. H. S. Moon, L. Lee, and J. B. Kim, "Double resonance optical pumping effects in electromagnetically induced transparency," *Opt. Express* **16**(16), 12163–12170 (2008).
10. H.-R. Noh and H. S. Moon, "Diagrammatic analysis of multiphoton processes in a ladder-type three-level atomic system," *Phys. Rev. A* **84**(5), 053827 (2011).
11. H. S. Moon and H.-R. Noh, "Resonant two-photon absorption and electromagnetically induced transparency in open ladder-type atomic system," *Opt. Express* **21**(6), 7447–7455 (2013).
12. H. Y. Ling, Y.-Q. Li, and M. Xiao, "Coherent population trapping and electromagnetically induced transparency in multi-Zeeman-sublevel atoms," *Phys. Rev. A* **53**(2), 1014–1026 (1996).
13. A. Godone, F. Levi, and J. Vanier, "Coherent microwave emission in cesium under coherent population trapping," *Phys. Rev. A* **59**(1), R12–R15 (1999).
14. L. V. Hau, S. E. Harris, Z. Dutton, and C. H. Behroozi, "Light speed reduction to 17 metres per second in an ultracold atomic gas," *Nature* **397**(6720), 594–598 (1999).
15. M. M. Kash, V. A. Sautenkov, A. S. Zibrov, L. Hollberg, G. R. Welch, M. D. Lukin, Y. Rostovtsev, E. S. Fry, and M. O. Scully, "Ultraslow Group Velocity and Enhanced Nonlinear Optical Effects in a Coherently Driven Hot Atomic Gas," *Phys. Rev. Lett.* **82**(26), 5229–5233 (1999).

16. R. Unanyan, M. Fleischhauer, B. W. Shore, and K. Bergmann, "Robust creation and phase-sensitive probing of superposition states via stimulated Raman adiabatic passage (STIRAP) with degenerate dark states," *Opt. Commun.* **155**(1), 144–154 (1998).
17. J. Vanier, "Atomic clocks based on coherent population trapping: a review," *J. Appl. Phys. B* **81**(4), 421–442 (2005).
18. D. Budker and M. V. Romalis, "Optical magnetometry," *Nat. Phys.* **3**(4), 227–234 (2007).
19. A. I. Lvovsky, B. C. Sanders, and W. Tittel, "Optical quantum memory," *Nat. Photonics* **3**(12), 706–714 (2009).
20. C. Y. Ye, A. S. Zibrov, Y. V. Rostovtsev, A. B. Matsko, and M. O. Scully, "Three-photon electromagnetically induced absorption and transparency in an inhomogeneously broadened atomic vapour," *J. Mod. Opt.* **49**(14), 2485–2499 (2002).
21. C. Carr, M. Tanasittikosol, A. Sargsyan, D. Sarkisyan, C. S. Adams, and K. J. Weatherill, "Three-photon electromagnetically induced transparency using Rydberg states," *Opt. Lett.* **37**(18), 3858–3860 (2012).
22. H. S. Moon and T. Jeong, "Three-photon electromagnetically induced absorption in a ladder-type atomic system," *Phys. Rev. A* **89**(3), 033822 (2014).
23. S. E. Harris, J. E. Field, and A. Imamoglu, "Nonlinear optical processes using electromagnetically induced transparency," *Phys. Rev. Lett.* **64**(10), 1107–1110 (1990).
24. J. C. Petch, C. H. Keitel, P. L. Knight, and J. P. Marangos, "Role of electromagnetically induced transparency in resonant four-wave-mixing schemes," *Phys. Rev. A* **53**(1), 543–561 (1996).
25. Y.-S. Lee and H. S. Moon, "Atomic coherence effects in four-wave mixing process of a ladder-type atomic system," *Opt. Express* **24**(10), 10723–10732 (2016).
26. Y.-Q. Li and M. Xiao, "Enhancement of nondegenerate four-wave mixing based on electromagnetically induced transparency in rubidium atoms," *Opt. Lett.* **21**(14), 1064–1066 (1996).
27. A. S. Zibrov, M. D. Lukin, and M. O. Scully, "Nondegenerate parametric self-oscillation via multiwave mixing in coherent atomic media," *Phys. Rev. Lett.* **83**(20), 4049–4052 (1999).
28. Y. Zhang, U. Khadka, B. Anderson, and M. Xiao, "Controlling four-wave and six-wave mixing processes in multilevel atomic systems," *Appl. Phys. Lett.* **91**(22), 221108 (2007).
29. F. E. Becerra, R. T. Willis, S. L. Rolston, and L. A. Orozco, "Nondegenerate four-wave mixing in rubidium vapor: the diamond configuration," *Phys. Rev. A* **78**(1), 013834 (2008).
30. R. T. Willis, F. E. Becerra, L. A. Orozco, and S. L. Rolston, "Four-wave mixing in the diamond configuration in an atomic vapor," *Phys. Rev. A* **79**(3), 033814 (2009).
31. U. Khadka, H. Zheng, and M. Xiao, "Four-wave-mixing between the upper excited states in a ladder-type atomic configuration," *Opt. Express* **20**(6), 6204–6214 (2012).
32. F. Wen, H. Zheng, X. Xue, H. Chen, J. Song, and Y. Zhang, "Electromagnetically induced transparency-assisted four-wave mixing process in the diamond-type four-level atomic system," *Opt. Mater.* **37**, 724–726 (2014).
33. S. Du, P. Kolchin, C. Belthangady, G. Y. Yin, and S. E. Harris, "Subnatural linewidth biphotons with controllable temporal length," *Phys. Rev. Lett.* **100**(18), 183603 (2008).
34. Y.-W. Cho, K.-K. Park, J.-C. Lee, and Y.-H. Kim, "Engineering frequency-time quantum correlation of narrow-band biphotons from cold atoms," *Phys. Rev. Lett.* **113**(6), 063602 (2014).
35. L. Zhao, X. Guo, Y. Sun, Y. Su, M. M. T. Loy, and S. Du, "Shaping the Biphoton Temporal Waveform with Spatial Light Modulation," *Phys. Rev. Lett.* **115**(19), 193601 (2015).
36. Y.-S. Lee, H.-R. Noh, and H. S. Moon, "Relationship between two- and three-photon coherence in a ladder-type atomic system," *Opt. Express* **23**(3), 2999–3009 (2015).
37. H.-R. Noh and H. S. Moon, "Three-photon coherence in a ladder-type atomic system," *Phys. Rev. A* **92**(1), 013807 (2015).
38. Y.-S. Lee and H. S. Moon, "Condition for Doppler-free three-photon resonance in a ladder-type atomic system," *Opt. Express* **23**(24), 31574–31581 (2015).

1. Introduction

Atomic coherence in multi-level atomic systems with various configurations induces interesting phenomena due to quantum interference [1–27]. In particular, two-photon coherence has long been studied in various atomic systems, such as Λ , V, and ladder configurations [1–11], and has been interpreted as quantum interference due to the interaction of two coherent fields and a three-level atomic system [8]. Coherent superposition between different states may lead to dramatic changes in the light absorption and propagation. Electromagnetically induced transparency (EIT), coherent population trapping (CPT), slow light, and stimulated Raman adiabatic passage (STIRAP) are well-known phenomena associated with two-photon coherence in a three-level atomic system [1–16]. Furthermore, two-photon coherence effects have been intensively studied in various applications, such as atomic clocks, atomic magnetometers, and quantum memory [17–19].

In contrast, in the context of a four-level atomic system interacting with three coherent fields, three-photon coherence has been studied in N, double- Λ , diamond, and cascade

configurations [20–27]. The following phenomena associated with three-photon coherence have been reported: three-photon electromagnetically induced transparency (TPEIT) [20, 21]; three-photon electromagnetically induced absorption (TPEIA) [20, 22]; and enhanced nonlinear optical processes [23]. Note that three-photon coherence phenomena based on a four-level atomic system have received relatively little attention compared with two-photon coherence. However, since the possibility of nonlinear optical process enhancement using EIT was first introduced by Harris *et al.* in 1990 [23], this enhanced four-wave mixing (FWM) has been intensively studied in various four-level atomic systems [24–32]. The EIT effect in resonant FWM has been investigated in order to elucidate the role of two-photon coherence in the FWM process [24]. For example, in a four-level diamond configuration, the relationship between two-photon coherence and FWM has been investigated under conditions of EIT and two-photon absorption [25]. Although the phenomena due to three-photon coherence differ from those due to two-photon coherence, the three-photon coherence effects are strongly correlated with the two-photon coherence. In particular, the spontaneous FWM process in four-level atomic systems has been intensively studied with regard to the interesting application of photon-pair generation, using a long atomic coherence between two ground states [33–35].

Recently, TPEIA has been both experimentally and theoretically investigated in a four-level atomic system composed of ladder and Λ configurations [22]. As revealed in earlier studies [6], the observation of TPEIA has also clearly shown that an atomic medium can be absorptive with a sub-natural width [36–38]. The enhanced absorption arises from three-photon coherence and results from constructive quantum interference [37]. Interestingly, it is possible to simultaneously observe both TPEIA and FWM in a four-level atomic system. However, the relationship between TPEIA and FWM in a four-level atomic system from the perspective of atomic coherence has not been investigated in detail.

In this paper, we investigate three-photon coherence in the four-level atomic system of the $5S_{1/2}$ – $5P_{3/2}$ – $5D_{5/2}$ transition of the ^{87}Rb atom. In particular, the TPEIA and FWM spectra are simultaneously observed and analyzed. To investigate the relationship between the TPEIA and FWM, we compare the spectral features of TPEIA and FWM signals according to the laser intensities and the laser frequency detuning. In addition, both signals are numerically calculated using a Doppler-broadened four-level atomic model, to further elucidate our experimental results for the TPEIA and FWM.

2. Four-level atomic system and experimental setup

Figure 1 shows the interaction of three coherent fields and a simple four-level atomic model composed of a ground state ($|1\rangle$), two degenerate intermediate states ($|2\rangle$ and $|3\rangle$), and an excited state ($|4\rangle$). In this paper, we introduce two kinds of TPEIA schemes in the four-level diamond-type atomic model, as shown in Fig. 1(a) and (b). We refer to the configuration in Fig. 1(a) as “upper Λ -type TPEIA”. This configuration is composed of a combination of ladder and Λ configurations. Also, it is worthy to note that the TPEIA has a velocity-selective effect, because a certain atomic velocity group is satisfied by the three-photon resonance condition [37, 38]. In Fig. 1(a), the probe (Ω_p) field interacts with the $|1\rangle$ – $|2\rangle$ transition, and the counter-propagating coupling (Ω_c) and driving (Ω_d) fields interact with the $|2\rangle$ – $|4\rangle$ – $|3\rangle$ transition (Λ -type) between the intermediate and excited states. On the other hand, Fig. 1(b) shows the “lower V-type TPEIA,” which features a combination of ladder and V configurations. The transition interacting with the driving field is changed from the $|4\rangle$ – $|3\rangle$ to the $|1\rangle$ – $|3\rangle$ transition, and the co-propagating the probe and driving fields interact with the $|2\rangle$ – $|1\rangle$ – $|3\rangle$ transition (V-type) between the ground state and the two intermediate states.

The experimental configuration used in this study was similar to that employed in previous studies of TPEIA [22, 25]. In our experimental setup, a 1-cm-long vapor cell containing isotopically pure ^{87}Rb was used, which was housed in μ -metal chambers to shield against the Earth's magnetic field. The probe, coupling and driving fields were applied by two external-cavity diode lasers, which were operated independently at wavelengths of 780.2 nm and 775.8 nm. All three fields are linearly polarized, and the Ω_p polarization was perpendicular to both Ω_c and Ω_d polarizations. The linear polarization can be described in circularly polarized basis, and when we consider the polarizations of three fields with Zeeman sublevels of each energy level, we can effectively describe the ladder-type configuration as a diamond configuration [37].

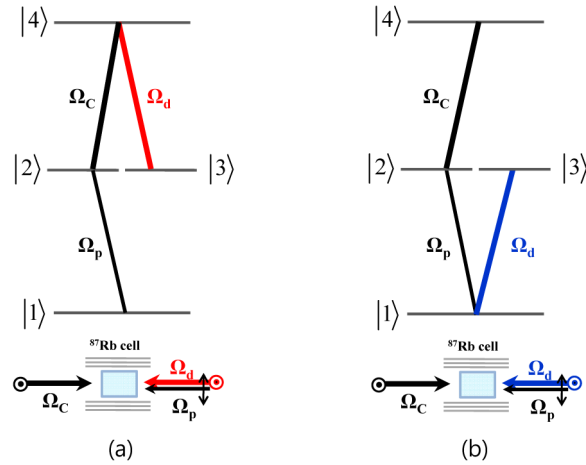


Fig. 1. (a) Upper Λ -type TPEIA and (b) lower V-type TPEIA configurations in simple four-level atomic model.

3. Experimental results and discussion

Figure 2(a) shows the energy-level diagram of the $5S_{1/2}$ – $5P_{3/2}$ – $5D_{5/2}$ transition of ^{87}Rb , for comparison of the EIT and the two kinds of TPEIA shown in Fig. 1(a) and (b). We can obtain ladder-type EIT (black curve) when there is no driving field, Λ -type TPEIA (red curve) when the driving field (red arrow) is coupled to the $5P_{3/2}$ – $5D_{5/2}$ transition, and V-type TPEIA (blue curve) when the Ω_d field (blue arrow) is coupled to the $5S_{1/2}$ – $5P_{3/2}$ transition, as shown in Fig. 2(a). Thus, the EIT, Λ -type TPEIA, and V-type TPEIA spectra as functions of the coupling field detuning frequency were observed depending on the driving field conditions in this study. The probe and driving frequencies were fixed at the $5S_{1/2}(F = 2)$ – $5P_{3/2}(F' = 3)$ transition, and that of coupling field was scanned over the $5P_{3/2}(F' = 3)$ – $5D_{5/2}(F'' = 2, 3, 4)$ transition. Transmittance spectra of the probe field are observed using a photo-diode [22]. We compared the EIT spectrum due to two-photon coherence with the two TPEIA spectra due to three-photon coherence, so as to analyze the relationship between the two- and three-photon coherences.

As shown by the black curve in Fig. 2(b), the ladder-type EIT spectrum in the $F = 2 - F' = 3 - F'' = 4$ cycling transition has a double structure composed of a narrow EIT component and a broad transmittance due to a saturation effect [9]. Double-resonance optical pumping (DROP), rather than the two-photon coherence, dominantly contributes to the transmittance signals in the $F'' = 2$ and 3 states. Note that the DROP effect in a ladder-type atomic system is not due to pure two-photon coherence [10].

Especially for the Λ -type TPEIA (red curve), this configuration exhibits the velocity-selective effect; a certain atomic group with a Doppler-shifted detuning satisfies the three-photon resonance condition. In this experiment, the narrow TPEIA was observed under the

condition of one-photon resonance for all three optical fields in a unique atomic velocity group. In Fig. 2(b), we clearly observe TPEIA peaks in the $F'' = 3$ and 4 states, because the TPEIA is related to the two-photon coherence. Comparing the hyperfine structure of the TPEIA with the EIT spectrum, it is apparent that the relative magnitude of the former is strongly related to that of the narrow EIT component, but it is not related to the DROP components [36].

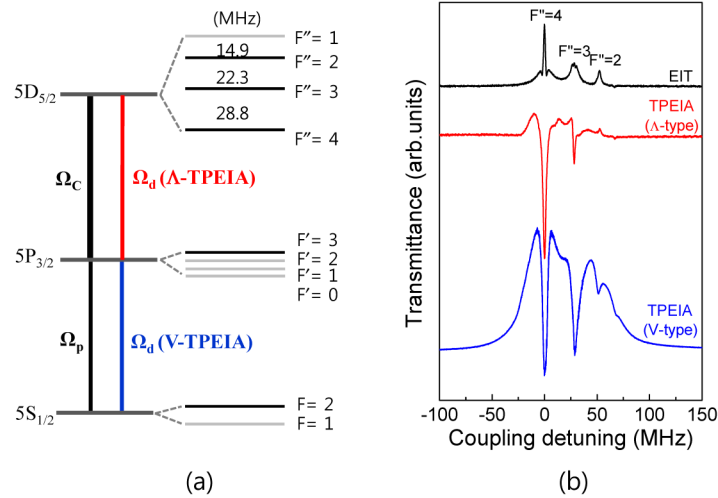


Fig. 2. (a) Energy-level diagram of $5S_{1/2}$ – $5P_{3/2}$ – $5D_{5/2}$ transition of ^{87}Rb , where the Ω_d fields for Λ - (red) and V- (blue) type TPEIAs are coupled to the $5P_{3/2}(F' = 3)$ – $5D_{5/2}(F'' = 4)$ and $5S_{1/2}(F' = 2)$ – $5P_{3/2}(F' = 3)$ transitions, respectively. (b) EIT (black curve), Λ -type TPEIA (red curve), and V-type TPEIA (blue curve) spectra as functions of Ω_c -field detuning frequency.

However, when the driving field was coupled to the $5S_{1/2}$ – $5P_{3/2}$ transition (blue arrow; Fig. 2(a)), the V-type TPEIA spectrum represented by the blue curve in Fig. 2(b) was observed. Note that, to the best of our knowledge, V-type TPEIA has never been investigated. The Doppler-free V-type TPEIA configuration can only be achieved when the probe and driving fields are co-propagating and have similar frequencies. When we denote the detuning frequencies of the probe, coupling and driving fields to δ_p , δ_c , and δ_d , the three photon resonance condition can be defined by $\delta_p + \delta_c - \delta_d = 0$. In the Doppler-free V-type TPEIA configuration, when we look at the two-photon resonance conditions for each V-type ($\delta_p - \delta_d = 0$) and ladder-type ($\delta_p + \delta_c = 0$) system, both two-photon resonance conditions are always satisfied in a wide range of Maxwell-Boltzmann distribution, which mean that three-photon resonance condition is satisfied. Therefore, all of the Doppler-broadened atoms satisfy the condition for three-photon resonance.

Below, we investigate the V-type TPEIA spectra obtained for different detuning frequencies of the three coherent optical fields in the Doppler-broadened medium. Comparing the V-type with the Λ -type TPEIA, interesting differences in the spectral features are observed. First, a broad transmittance background exists for the V-type TPEIA, having a spectral width of approximately 50 MHz and resulting from the saturation effect at the cycling transition and the optical pumping effects. Second, the three hyperfine structures of the V-type TPEIA configuration can be clearly observed at the $5D_{5/2}(F'' = 2, 3, \text{ and } 4)$ states. This difference can be understood as resulting from the optical pumping effect due to the strong driving field. The causes of this optical pumping are the DROP and the two-step excitation with the strong coupling and driving fields, including the $5D_{5/2}$ – $6P_{3/2}$ transition. We discuss the TPEIA spectra according to the powers of the driving and coupling fields in detail below.

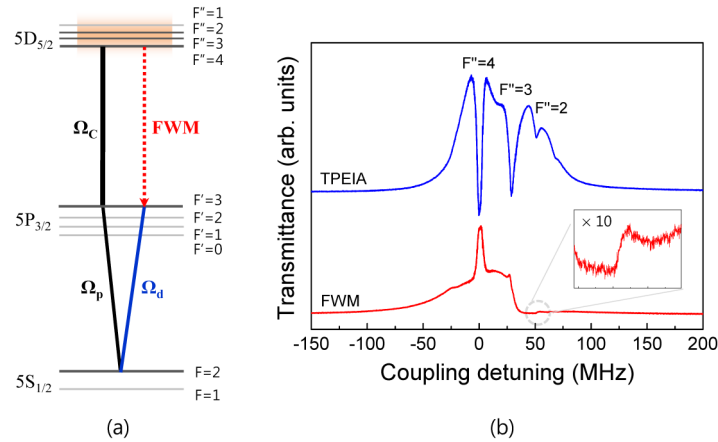


Fig. 3. (a) V-type TPEIA and FWM configuration for $5S_{1/2}$ - $5P_{3/2}$ - $5D_{5/2}$ transition of ^{87}Rb . (b) V-type TPEIA (blue curve) and FWM (red curve) spectra as functions of Ω_c -field detuning frequency.

As our previous studies revealed not only the relationship between two- and three-photon coherence [36] but also various two-photon coherence effects to FWM [25], it is natural to look into the relationship between three-photon coherence and FWM from a view point of atomic coherences. As mentioned in the Introduction, a four-level atomic system interacting with three coherent fields is important not only for TPEIA, but also for FWM signals. For the configuration shown in Fig. 3(a), it is possible to observe V-type TPEIA and FWM spectra simultaneously. The TPEIA due to the three-photon atomic coherence effect should be satisfied with both ladder-type (interacting with the probe and coupling fields) and V-type (interacting with the probe and driving fields) two-photon coherences. Furthermore, the FWM process is significantly influenced by both two-photon coherences [24, 25]. The FWM light is stimulated via interaction of the Ω_d field with the atomic medium in the presence of two-photon coherence. Thus, the FWM process can be understood as three-photon interaction involving the probe and coupling fields for two-photon coherence and the driving field for the stimulated process. Therefore, we can assume that the FWM signal is strongly correlated with the TPEIA as a result of the three-photon coherence.

In Fig. 3(b), we compare the FWM (red curve) with the V-type TPEIA (blue curve) spectra. In this experiment, the powers of the probe, coupling, and driving fields were 0.2 mW, 5 mW, and 50 mW, respectively, and the beam diameters of the three beams were 2.2 mm. The intensities of the three fields were estimated as 5.3 mW/cm², 132 mW/cm², and 1315 mW/cm², respectively. The direction of the generated FWM light was counter-propagated to that of the driving field under a phase matching condition. Further, the FWM light was linearly polarized perpendicular to that of the driving field. For the $F = 2 - F' = 3 - F'' = 4$ cycling transition, the FWM spectral shape has a double structure with both a narrow and a broad FWM signal of approximately 50 MHz width. The TPEIA spectral shape in this cycling transition is comprised of a narrow TPEIA signal due to three-photon coherence and a broad transmittance background due to the saturation effect. Here, the saturation effect includes two-photon coherence with population changes between the $5S_{1/2}$ and $5D_{5/2}$ states. From both spectra in Fig. 3(b), it is apparent that the narrow FWM signal is strongly correlated with the three-photon coherence of the narrow TPEIA signal. However, in the case of the $F = 2 - F' = 3 - F'' = 2$ and 3 open transitions, the FWM spectral shape differs significantly from that of the TPEIA. FWM does not show a broad transmittance signal while TPEIA has in the open transition. This is because FWM is only induced by three-photon coherence which is related to pure two-photon coherence, whereas the TPEIA signal always contains the strong saturation effect since we have one intermediate state resonating to strong driving and

coupling field. If the two intermediate states have different energy levels, the coupling field is not resonant to the driving field, so that the saturation effect due to driving field will be significantly reduced. From the observed TPEIA and FWM spectra shown in Fig. 3(b), we assume that the FWM light generation is partially due to the enhanced TPEIA absorption signal. We investigate the relationship between the TPEIA and FWM according to the laser frequency detuning and the laser intensities in detail below.

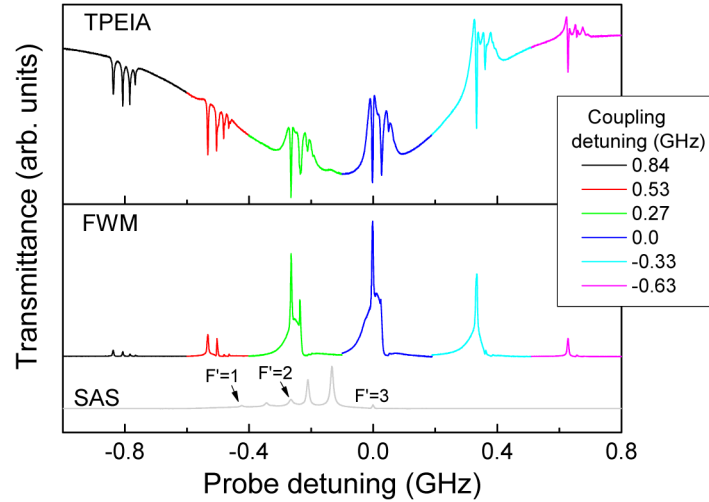


Fig. 4. TPEIA and FWM spectra according to Ω_C -field detuning frequency, where the x-axis is the relative detuning frequency of the Ω_p field; the gray curve is the saturated absorption spectrum (SAS) of the Ω_p field.

We investigated the TPEIA and FWM spectra according to the Ω_C -field detuning frequencies in the Doppler-broadened medium, as shown in Fig. 4. The frequency of the Ω_p field was detuned from -1.0 GHz to 0.8 GHz at the $F = 2 - F' = 3$ transition, and TPEIA and FWM signals were simultaneously obtained for six different Ω_C -field detuning frequencies (0.84 GHz, 0.53 GHz, 0.27 GHz, 0 , -0.33 GHz, and -0.63 GHz). Also, the detuning frequency of probe field is the same as that of driving field, since the two fields share one laser. So, the broad transmittance signal still appears in the detuned frequency of probe field. The gray curve in Fig. 4 shows the saturated absorption spectrum (SAS) of the Ω_p field, which is used to mark the optical frequency. The TPEIA spectra have the Doppler background and dynamically change according to the Ω_C -field detuning frequency. However, although the FWM signals did not exhibit a Doppler background, the entire magnitude of the FWM changed with the Doppler profile according to the Ω_C -field detuning frequency. Comparing the TPEIA and FWM spectra, it is apparent that both signals are closely correlated with each other.

For the V-type TPEIA configuration of Fig. 1(b), the TPEIA and FWM signals were continuously observed under detuning free of the Ω_C field. The ladder-type (interacting with the counter-propagating Ω_p and Ω_C fields) and V-type (interacting with the co-propagating Ω_p and Ω_d fields) two-photon coherences were Doppler-free in the Doppler-broadened medium. Thus, the Doppler-free condition of both the ladder and V-type two-photon resonances should be satisfied in order to maintain the three-photon atomic coherence effect due to quantum interference in the Doppler-broadened medium. As shown in Fig. 4, Doppler-free TPEIA was clearly observed over the entire Doppler broadened range.

As the detuning frequencies from the $F = 2 - F' = 3$ transition increased, the broad FWM signal of the cycling transition was removed and the broad transmittance in the TPEIA background decreased. This result can be understood as indicating that the saturation effect

and the population change due to single-photon resonance decrease significantly under far detuning conditions. Therefore, under the condition of far detuning from the Doppler-broadening and for detuning frequencies of -0.8 GHz, -0.5 GHz, and 0.6 GHz, the both TPEIA and FWM signals show the only narrow spectral features without broad transmittance which is due to pure three-photon coherence, as shown in Fig. 4.

The FWM signal of the $F'' = 3$ state was obtained under the broad FWM signal of the cycling transition. However, as the Ω_p -field frequency is further red-detuned, the FWM peak of the $F'' = 3$ state was clearly observed, as shown in Fig. 4. For a far detuning frequency of -0.8 GHz, the hyperfine structures of the $5D_{5/2}(F'' = 1, 2, 3, 4)$ states in both the TPEIA and FWM spectra can be seen. As shown in the inset of Fig. 4, the magnified FWM spectrum clearly resolves the hyperfine structure of FWM from which one may figure out its relation to two-photon coherence [25]. Although the FWM is not completely proportional to the TPEIA, the magnitudes and spectral features of both signals are correlated with each other. In this paper, we interpret the FWM as one of three photon coherence phenomena. However, it is also hard to claim that the absorption signal at a far detuning frequency of -0.8 GHz is fully due to three-photon coherence. The absorption could be dominantly affected by two-photon coherence at far detuned regime because the number of atom contributing to three-photon coherence is small at the edge of Maxwell-Boltzmann distribution.

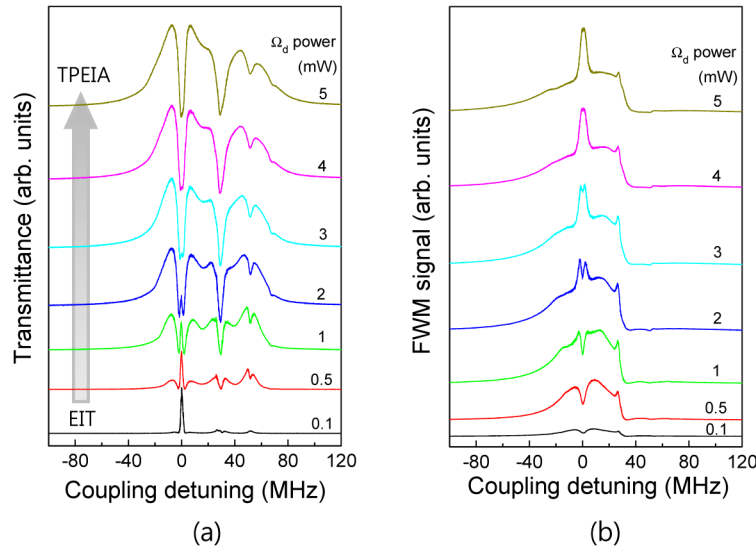


Fig. 5. (a) TPEIA and (b) FWM spectra according to Ω_d power.

To investigate the relationship between the TPEIA and FWM in terms of the three-photon coherence, TPEIA and FWM spectra were obtained for different Ω_d powers, and for weak Ω_p and strong Ω_c powers of 0.2 mW and 50 mW, respectively, as shown in Fig. 5(a) and (b). As the Ω_d power increased from 0 to 5 mW, the Ω_p field transmittance in the cycling transition changed from EIT due to two-photon coherence to TPEIA due to three-photon coherence. Comparing the TPEIA and FWM spectral shapes in the cycling transition, it is clear that the spectral features of the FWM signals according to the Ω_d power switched from dips to peaks, apparently reflecting the opposite behavior in the TPEIA spectra. Hence, we confirmed that the TPEIA is strongly correlated with the FWM, because of the three-photon coherence.

In addition, as the Ω_d power increased, the TPEIA background transmittance increased and broadened, as shown in Fig. 5. As mentioned above, this transmittance is due to the saturation effect in the cycling transition and the DROP effect in the $F'' = 2$ and 3 states. Comparing the TPEIA and FWM, we may assume that the saturation effect contributed to the

FWM signal, but the DROP effect did not. The saturation effect in the cycling transition causes transfer of the $F = 2$ ground state population to the $F'' = 4$ excited state, through the $F' = 3$ intermediate state. The atoms in the $F'' = 4$ excited state can contribute to the FWM signal. However, the DROP effect in the open transitions means that the population of one ground state ($F = 2$) may be depleted, because the atoms of the $F = 2$ state can be optically pumped into another ground state $F = 1$ through the $F' = 1$ and 2 intermediate states. Because the Ω_d -field frequency is resonant on the $F = 2 - F' = 3$ transition, the increase in the Ω_d power changes the population significantly due to the saturation and DROP effects. Therefore, we can understand the relationship between the TPEIA and FWM with regard to the broad spectral feature based on the saturation and DROP effects.

4. Theoretical analysis

To elucidate the relationship between the TPEIA and FWM, we theoretically investigated these phenomena in the simple four-level atomic system of Fig. 1(a). From the density matrix equations [22], we obtained the steady-state coherence components related to the TPEIA and the FWM, ρ_{12} and ρ_{34} , respectively, which are proportional to $\text{Im}(\rho_{12})$ and $|\rho_{34}|^2$, respectively. Under weak Ω_p conditions, ρ_{12} and ρ_{34} can be expressed as

$$\rho_{12} = \frac{\Omega_c \rho_{14} - \Omega_d \rho_{23}}{2\delta_p - i\Gamma_{12}}, \quad (1)$$

and

$$\rho_{34} = \frac{\Omega_d \rho_{14} - \Omega_c \rho_{23}}{2(\delta_c + \delta_p - \delta_d) - i(\Gamma_{13} + \Gamma_{34})}. \quad (2)$$

Here, the detuning frequencies of Ω_p , Ω_d , and Ω_c are δ_p , δ_d , and δ_c , respectively, and the decay rate between the $|i\rangle$ and $|j\rangle$ states is represented by Γ_{ij} .

Interestingly, it is apparent that both ρ_{12} and ρ_{34} of Eqs. (1) and (2), respectively, are directly related to the ladder- and V-type two-photon coherence components ρ_{14} and ρ_{23} , respectively. When the Rabi frequencies of both Ω_d and Ω_c are equal, the relationship between ρ_{12} and ρ_{34} is proportional, because the numerator term of ρ_{12} is equal to that of ρ_{34} . Furthermore, under the conditions of $\Omega_d = \Omega_c$ and $\delta_d = \delta_c$, we can obtain the simple relation $\rho_{12} \approx \rho_{34}$, because the Γ_{34} decay (0.6 MHz) is ten times smaller than that of Γ_{12} or Γ_{13} (6.0 MHz). Therefore, as assumed above, we can easily understand that the FWM is strongly correlated with the TPEIA due to three-photon coherence.

We numerically investigated the change in the FWM signals during the probe transmittance signal transformation from EIT to TPEIA in response to the additional Ω_d field, which is similar to the experimental results for the $F = 2 - F' = 3 - F'' = 4$ cycling transition shown in Fig. 5. Figures 6(a) and (b) show $-\text{Im}(\rho_{12})$ for V-type TPEIA and $|\rho_{34}|^2$ for FWM according to Ω_d , which were obtained by applying the density matrix equation to the four-level atomic system of Fig. 1(b). The calculated V-type TPEIA and FWM spectra were considered for the Doppler-broadened atomic medium and the branching ratios for the cycling transition, where the Rabi frequencies of Ω_p and Ω_c were set to 1 MHz and 15 MHz, respectively. When Ω_d was increased from 0 to 15 MHz, the calculated spectra changed from EIT due to two-photon coherence to TPEIA due to three-photon coherence, as shown in Fig. 6(a). Comparing the FWM spectral shapes in Fig. 6(b) with the calculated spectra of Fig. 6(a), it is clear that the calculated spectra switched from EIT to TPEIA, with this behavior

appearing to be oppositely reflected in the $|\rho_{34}|^2$ spectra. Of course, the simple four-level atomic model cannot be considered to exhibit the hyperfine structures and Zeeman sublevels of a real atomic system; however, we successfully employed this approach to elucidate the three-photon coherence effects of the simultaneously observed TPEIA and FWM of Fig. 5 based on the calculated spectra of Fig. 6.

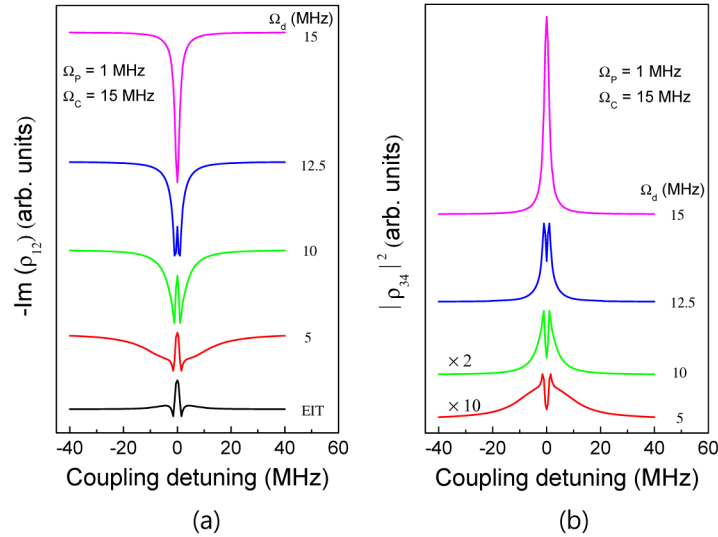


Fig. 6. Numerically calculated (a) V-type TPEIA and (b) FWM spectra (black curves) according to Rabi frequency (Ω_d) of driving field.

5. Conclusion

We simultaneously observed TPEIA and FWM phenomena due to three-photon coherence effects and investigated the relationship between both of these phenomena in a four-level configuration of the $5S_{1/2}$ – $5P_{3/2}$ – $5D_{5/2}$ transition of the ^{87}Rb atom. In this work, we introduced the lower V-type TPEIA scheme, and compared it with the upper Λ -type scheme. The lower V-type TPEIA was obtained for three-photon resonance in the Doppler-broadened atomic medium, because of the Doppler-free configuration among the three optical fields. When we investigated the relationship between the lower V-type TPEIA and FWM spectra by simultaneously detecting the probe and FWM fields, we found similarities in the spectral shapes of the FWM and the TPEIA induced by the three-photon coherence only, excluding the optical effects. Although the magnitude of the FWM spectrum is not completely proportional to that of the TPEIA spectrum, we confirmed that the magnitudes and spectral features of both signals are closely correlated with each other and that the FWM light emission is related to the enhanced TPEIA absorption signal. As the driving-field power increased, the ladder-type EIT signal of the probe field was transformed to TPEIA and the FWM signals switched from dips to peaks. Hence, we suggested that the TPEIA and FWM phenomena are interpreted as one of the three-photon coherence phenomena induced by interaction with the driving field. From theoretical results obtained for the Doppler-broadened four-level atomic model, we clarified that the FWM is strongly correlated with the TPEIA due to three-photon coherence. Our results contribute to further understanding of the three-photon coherence effects in four-level atomic systems interacting with three coherent fields. In future, we hope to contribute to a better understanding of the properties of photon pairs obtained via a spontaneous four-wave mixing process in a four-level atomic system.

Funding

National Research Foundation of Korea (2015R1A2A1A05001819).